AI for Good Sandbox Hackathon – ShifaMiko – MEDIMCOM IA

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Github link: https://github.com/baudouin2/MEDIMCOM-IA

1. **Introduction**

   Deaf patients, especially those who read little, often cannot describe their symptoms or understand a treatment without a sign-language interpreter, and interpreters are scarce, above all in emergencies. Critical facts get lost (pregnancy, allergies, current medicines), consent is not truly informed and instructions are misunderstood. **MEDIMCOM IA** is an agentic AI solution that lets a doctor and a deaf patient communicate through pictograms, while autonomous agents keep the patient record reliable and check every prescription, every message to the patient and every data request against UAE law and the standards of the health regulators (DoH Abu Dhabi, DHA, MOHAP). The doctor always decides.

   This submission maps the technical solution to the policies that frame it. Each node of the ITU-T Y.3172 pipeline is linked to the UAE laws and standards that govern it and to the dimensions of the ITU AI Readiness Report 2.0, including AI for inclusion, human interface, AI and policies, and collaboration with AI. The knowledge base built for the agents (47 sources, 139 provisions, each with its official link and verification level) is published with the code. Where no binding text settles a situation, the solution records a policy or standards gap; nine are reported in section 6. The prototype uses fictitious patients only and is not a medical device.

2. **Description of the use case**

   Existing solutions and their gaps. Today a deaf patient relies on a sign-language interpreter, rarely available at short notice and often absent at night or in emergencies; on written notes, which fail with patients who read little; on relatives, who break confidentiality and filter information; or on generic translation and chat tools, which are not designed for clinical safety, can invent content and may send health data abroad, which UAE Federal Law No. 2 of 2019 on ICT in Health (Art. 13) forbids. None of them checks whether what is said or prescribed is safe and lawful, and none leaves an auditable record.

   Our solution. A local web application with two screens (the doctor's screen and a patient tablet showing pictograms only, with optional Arabic, English or French labels and an emergency button) and three autonomous agents that share a versioned consultation record:

   - **Translator agent:** turns the doctor's questions and validated prescriptions into pictograms, and the patient's taps back into text. When no safe pictogram exists for a word, it alerts the doctor instead of guessing.
   - **Assistant agent:** keeps the patient state (confirmed and unconfirmed facts), spots missing information (e.g. pregnancy status before a prescription), proposes questions and sourced treatment options, and reads the documents the patient sends (photos of earlier prescriptions or receipts). It never talks to the patient and never prescribes.

- **Jurist agent:** checks each prescription, message and data request with 81 deterministic rules drawn from the knowledge base, and gives one of four verdicts, each citing the provisions applied: COMPLIANT; INSUFFICIENT EVIDENCE (an unknown fact is never assumed safe); REVIEW (the doctor may proceed with a written justification logged under his or her name); BLOCKED (no override). Only a binding text can block; a clinical recommendation can at most require review.
- **Institutional agents (crisis):** a health authority, a reference hospital and a crisis cell each receive only their own fragments of information, never see the patient record, and act only on a verified legal basis within their mandate. The jurist answers their legal questions.

How it closes the gaps. Pictograms replace the missing interpreter; the jurist adds the safety and legal check that other tools lack; processing runs on the clinic computer, so health data stay in the country (an online model can be plugged in for tests with fictitious data only); and every reasoning step, message, verdict and human decision is written to a tamper-evident, hash-chained journal. The agents are autonomous: each one observes the shared record, reasons and chooses which of its tools to use. For example, the jurist re-checks a prescription on its own when a new answer from the patient changes the facts. Their limits are enforced in their tools, not only in their instructions.

3. **Use case requirement**

Each node of the pipeline carries requirements derived from the UAE texts in the knowledge base:

- **SRC:** the patient must be able to give every fact needed for a safe prescription without reading or writing (symptoms, pain level, pregnancy, allergies, current treatments); the doctor may type or dictate; this prototype uses fictitious patients only.
- **C:** data are collected on a local server in the facility; no health data are stored or processed outside the UAE (Law No. 2 of 2019, Art. 13); patient documents are limited in number and size.
- **PP:** a fact is marked confirmed only when the patient or the doctor confirmed it; facts read from documents stay unconfirmed until the doctor checks them; names and identifiers are removed before any message to an institution (Law No. 2 of 2019, Art. 16; PDPL, Decree-Law No. 45 of 2021).
- **M:** the language model only assists; legal verdicts must not depend on it; the system keeps working on its written rules if the model fails.
- **P:** a licensed human doctor validates every message and decides every prescription (Medical Liability Law, Decree-Law No. 4 of 2016; DoH and DHA AI policies); an unknown fact is never treated as safe; blocking requires a binding text.
- **D:** nothing reaches the patient without the doctor's validation and the jurist's check; consent and explanations are given in a form the patient understands (Law No. 29 of 2006, Art. 7; DHA Telehealth Standards).
- **SINK:** every step is traceable and tamper-evident (DoH Responsible AI Standard 2025, Sec. 3.1.7.1.3) and can be kept for the 25-year retention period of Law No. 2 of 2019, Art. 20.

4. **Mapping with ITU-T Y.3172, Readiness 2.0 Report, and domain-specific policies**

Node	Use case (MEDIMCOM IA, health care)	Readiness 2.0 Report	Documents (policies, strategies, regulations)
SRC	Doctor's typed or dictated text; patient's pictogram taps on a tablet; record form (age, weight, pregnancy, allergies, treatments); photos of earlier prescriptions sent by the patient; knowledge base (47 sources, 139 provisions)	Data · D1 data and model marketplace (open sources with links); D11 AI for inclusion (text-free input)	Law No. 2/2019 (ICT in Health); PDPL 45/2021; Law No. 29/2006 (people of determination); UN CRPD, Art. 21
C	Local server on the clinic computer; events with tap, session and time; at most 10 documents of 12 MB per patient, de-duplicated by hash; official legal texts downloaded with a frozen hash	Deployment support · D6 human interface (image-only screen, ar/en/fr labels); D13 digital infrastructure (runs offline on one laptop)	Law No. 2/2019, Art. 13 (data stay in the UAE); DHA Telehealth Standards v4 (identification, accessible channel)
PP	Typing and dictation errors corrected; doses normalised; facts kept as confirmed or unconfirmed; identifiers removed before messages to institutions; legal texts cut into 148 indexed chunks	Data · D10 AI and policies (data sovereignty: local processing)	PDPL 45/2021; MOHAP Resolution 51/2021 (transfers abroad); Law No. 2/2019, Art. 16 (secondary use)
M	Assistant: local open-source model or an online model for tests with fictitious data; knowledge-base retrieval (recall@3 = 0.89 on 65 test questions); jurist: 81 deterministic, formally verified rules	Research; Open source · D8 collaboration with AI (agents compared with a scripted baseline)	DoH Abu Dhabi Responsible AI Standard 2025; DHA AI in Healthcare Policy 2021
P	Jurist verdicts; doctor validates every message; REVIEW needs a named, written reason; BLOCKED cannot be overridden; institutions act only on a binding basis	Standards · D10 AI and policies (policy tested in 3 sandbox scenarios; 9 gaps)	Decree-Law 4/2016 (medical liability); DoH AI Policy 2018 (AI as support); UAE AI Charter 2024 (human oversight)
D	Pictogram messages sent only after validation and the jurist's check, each prescription with a consent question; typed messages between agents and institutions	D7 strategy alignment (institutions coordinate through their mandates)	DHA Telehealth Standards v4 (consent, explanation); Decree-Law 4/2016, Art. 5 (consent)
SINK	Doctor and patient screens; crisis board; hash-chained audit journal and readable responsibility report	D11 AI for inclusion (pictogram comprehension rate, ISO 9186-1 test planned); Sandbox (reproducible journal)	Law No. 14/2014 (communicable diseases, 24-hour notification); Law No. 2/2019, Art. 20 (retention)

Beyond data and models, the mapping covers strategy alignment (D7), interface design for inclusion (D6, D11) and human–AI collaboration (D8); all six readiness factors are addressed. Of the 47 knowledge-base sources, 33 were read on the official publisher's site, 11 come from reliable secondary sources and 3 remain to be confirmed; the level is shown in every citation, and non-binding guidelines are labelled as such. Official links are listed in the repository (KB_reference_links.md), and the base is packaged for the ITU reference ingestion code. A second country file (Cameroon) shows that the knowledge base can be swapped.

5. **Error scenarios**

Information is released fragment by fragment; each agent sees only its own messages.

Scenario 1 – Pregnancy and an unsafe medicine

1. Step-1: A deaf woman, 25, has abdominal pain; her pregnancy status is unknown. The assistant, on its own, asks “Are you pregnant?”; the translator turns it into pictograms; the doctor validates; the jurist clears it.

2. Step-2: Before the answer arrives, the doctor prescribes ibuprofen. The jurist answers INSUFFICIENT EVIDENCE (pregnancy not confirmed; Decree-Law 4/2016, Art. 4(2)) instead of assuming it is safe.

3. Step-3: The patient taps YES. The jurist re-checks the prescription without being asked (REVIEW: anti-inflammatory drug in pregnancy); the assistant proposes paracetamol with its source; the doctor prescribes it: COMPLIANT, with consent and accessible-information obligations.

Scenario 2 – Data and model under pressure

1. Step-1: A company asks for the list of pregnant patients: BLOCKED (Law No. 2/2019, Art. 16), with the lawful path stated.

2. Step-2: The local model fails and someone proposes sending the consultation to a foreign cloud AI: BLOCKED (Art. 13; fine under Art. 24). The assistant falls back on its written rules; the jurist never depended on the model.

3. Step-3: An external module tries to send a message to the patient without validation: BLOCKED; once the doctor validates the translator's version, it is sent.

Scenario 3 – Communicable-disease crisis

1. Step-1: Tramadol is prescribed by teleconsultation and the doctor insists: BLOCKED (DHA Telehealth Standards v4, Sec. 17.12.7); a blocking rule has no override.

2. Step-2: A notifiable disease is suspected. The jurist reminds the 24-hour notification (Law No. 14/2014, Art. 4). The health authority, on its own, opens an investigation, asks the jurist about its powers and declares an infected area (Art. 18). A foreign laboratory asks for identified results: REVIEW; the doctor authorises the listed exception by name.

3. Step-3: Only the crisis cell learns that the press wants the patients' names: BLOCKED (Art. 29); it publishes anonymised figures instead. Wrong pictograms are reported in three facilities: the cell logs an AI incident and asks the jurist where to report it (gap L08).

Where the solution faces controversy, and how it responds:

- **The doctor disagrees with the AI.** On REVIEW the doctor may proceed with a written reason logged under his or her name. A BLOCKED verdict cannot be overridden, which can be contested in urgent care: the legal emergency exception to consent is built in, and every disagreement is kept for audit.
- **Consent given through pictograms** has no clear legal status (gap L04): each explanation ends with a consent question, the patient can answer “I don't understand”, which alerts the doctor, and the gap is flagged instead of claiming valid consent.
- **Liability after a validated AI suggestion causes harm** (gap L01): the journal shows who proposed, what was checked and who decided, giving evidence that a regulator or court can use.

- **Online models and data sovereignty** (gap L06): an online model is used for tests with fictitious data only (reminded at every start-up); with real patients the solution is designed to run on local open-source models.

Evaluation. Each scenario is checked automatically: 29 capability and safety expectations and 6 safety invariants, all met in the reference runs, also with a random order of agent turns; 89 safety and utility properties of the UAE rules are proved for all inputs (exhaustive enumeration and SMT proofs). **Limits:** reference results use scripted agent policies; only 7 documented medicines are checked (others give INSUFFICIENT EVIDENCE); clinical rules must be validated by a physician; the pictograms have not yet been tested with deaf users.

6. **Policy gaps and standards gaps derived from the mapping exercise**

The jurist records a gap when no binding UAE text settles a situation it meets. Gaps are findings for the domain, illustrated by global examples, not recommendations to a single country. Details: Gap Analysis annex in the repository.

Gap	What is missing in UAE texts	Global example
L01 Liability for a validated AI suggestion	No rule sharing liability between clinician, developer and facility; no rule on the evidential value of AI logs	EU Product Liability Directive 2024/2853; EU AI Act, Art. 14
L02 When is a decision-support or communication assistant a medical device?	No criteria separating regulated decision-support software from communication aids and compliance checkers	FDA CDS guidance; EU MDR, Rule 11; MHRA AI Airlock
L03 Communication needs of deaf patients	No operational standard requiring providers to identify, record and meet a patient's communication need	NHS Accessible Information Standard (DCB1605)
L04 Consent given through pictograms	Consent through alternative communication not recognised; no rule on checking comprehension or documenting it	UN CRPD, Arts. 12, 21, 25
L05 Validation of medical pictograms	No standard on testing health pictograms or AI-generated images shown to patients	ISO 9186-1:2014
L06 Data localisation vs cloud AI and sandboxes	No pathway for cross-border AI sandboxes; status of synthetic or de-identified data unclear	EU AI Act, Arts. 57–59; MHRA AI Airlock
L07 Oversight of multi-agent systems	No rules on agent-to-agent communication, agent autonomy, tool permissions or tamper-evident logs	EU AI Act, Arts. 12, 14
L08 Reporting AI incidents across emirates	Duties differ by emirate; no federal time limit or shared format	EU AI Act, Art. 73; WHO IHR (2005), Art. 6
L09 Emergency access for deaf people	Emergency protocols do not require equivalent non-voice access	EU Electronic Communications Code, Art. 109(5)

These gaps can be proposed to the relevant ITU-T study groups; MEDIMCOM IA already implements a response to several of them (tamper-evident multi-agent journal, consent question with an “I don't understand” answer, local processing).